Exploring the EEG signals of Schizophrenia

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Background

The brain typically suppresses its response to expected stimuli (sensory gating).

Schizophrenia is associated with impaired sensory gating[1].

Figure 1: EEG illustration[2]

Data

Our data were taken from the National Institute of Mental Health (NIMH project number R01MH058262), available from Kaggle

In our data, 36 subjects were exposed to three conditions (25 controls, 11 cases):

1. Subjects press a button to immediately produce a sound
2. Subjects passively listen to the same sound without pressing a button
3. Subjects press a button, and no sound is produced.

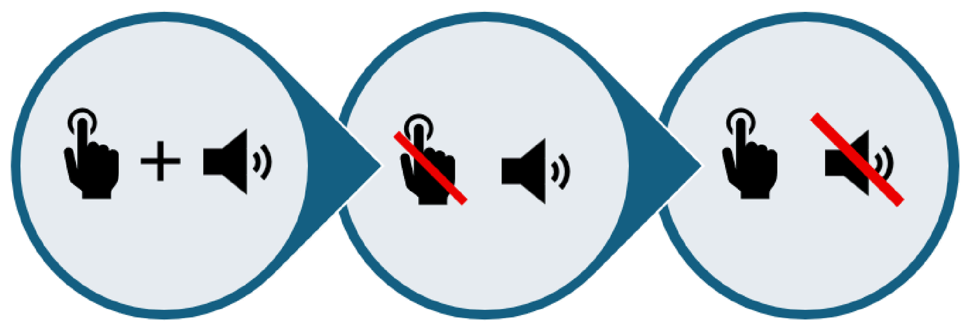

Figure 2: Condition Visualisation

Objectives of Project

- ***Hypothesis:*** Individuals with schizophrenia exhibit statistically different EEG signals compared to healthy controls during conditions ~100ms.
- ***How?:*** Functional Data Analysis (FDA) was applied to investigate the difference in EEG signals in the FP1 node.
- ***Why FP1?:*** Previous EEG studies reported altered frontal activity and reduced frontal coherence in relation to schizophrenia research. [3]

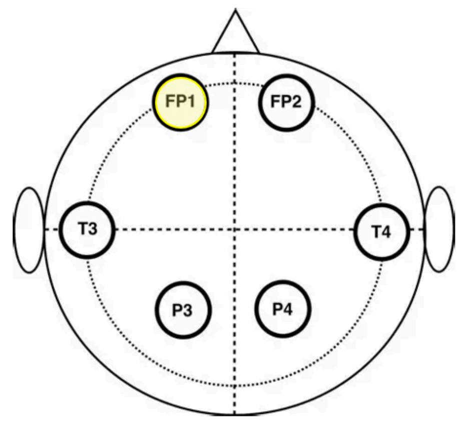

Figure 3: Nodes Cap[4]

Results

Raw Data

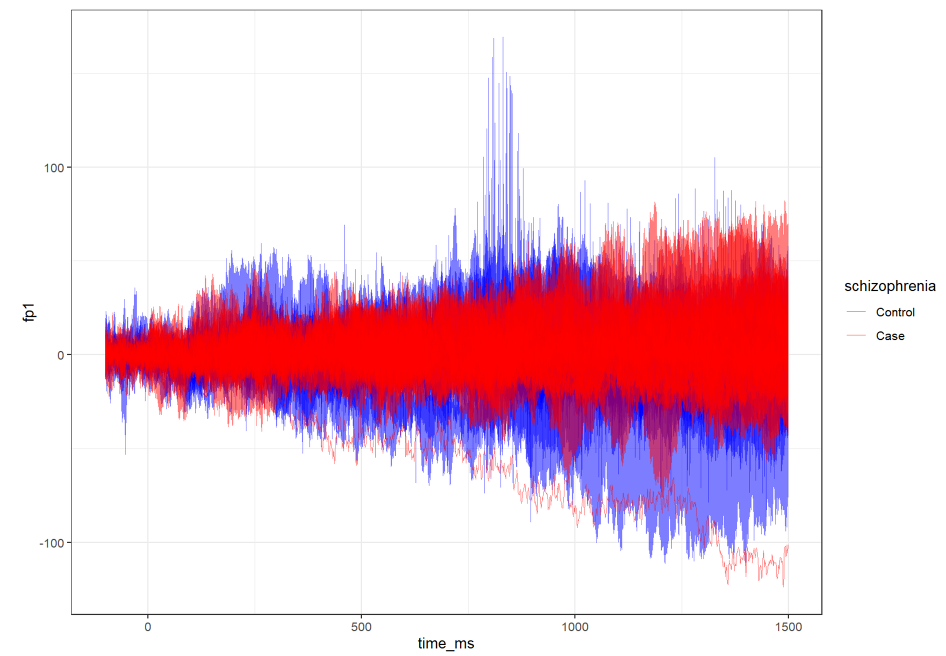

Figure 4: Raw Data Plot, Condition 1: Cases vs Controls

Smoothing Process

Spline smoothing was applied to remove noise from data:

- 100 cubic B-spline functions between [0, 1638]ms
- Second derivative penalty applied

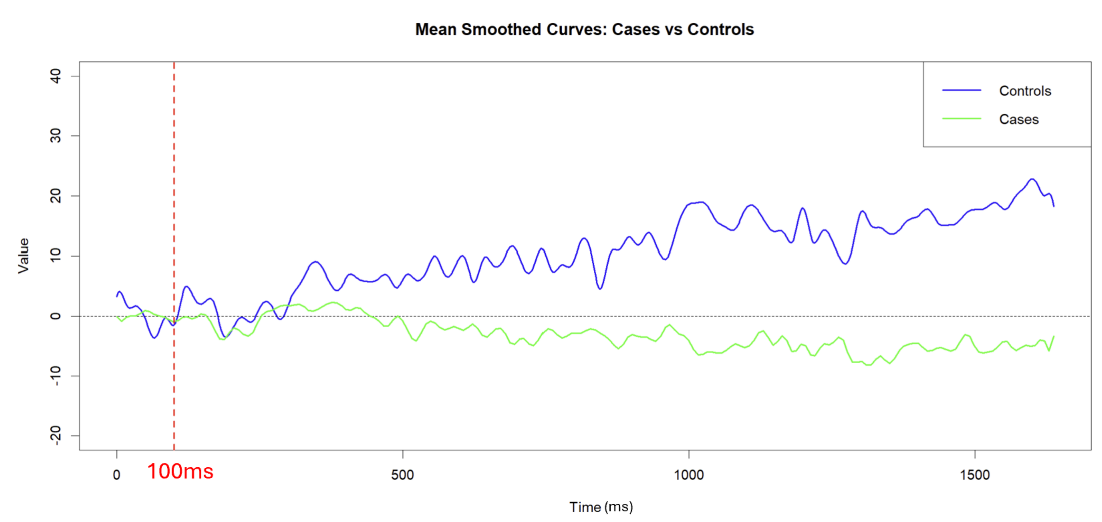

Figure 5: Mean Smoothed Curves Plot, Condition 1: Cases vs Controls

Permutation T-Test

- Testing H_0, no significant difference between cases and controls
- Re-sampling technique that generates a test statistic on each permutation
- Comparison of the permutation null distribution and observed data following random reshuffling. The observed difference signifies evidence of a difference between the groups

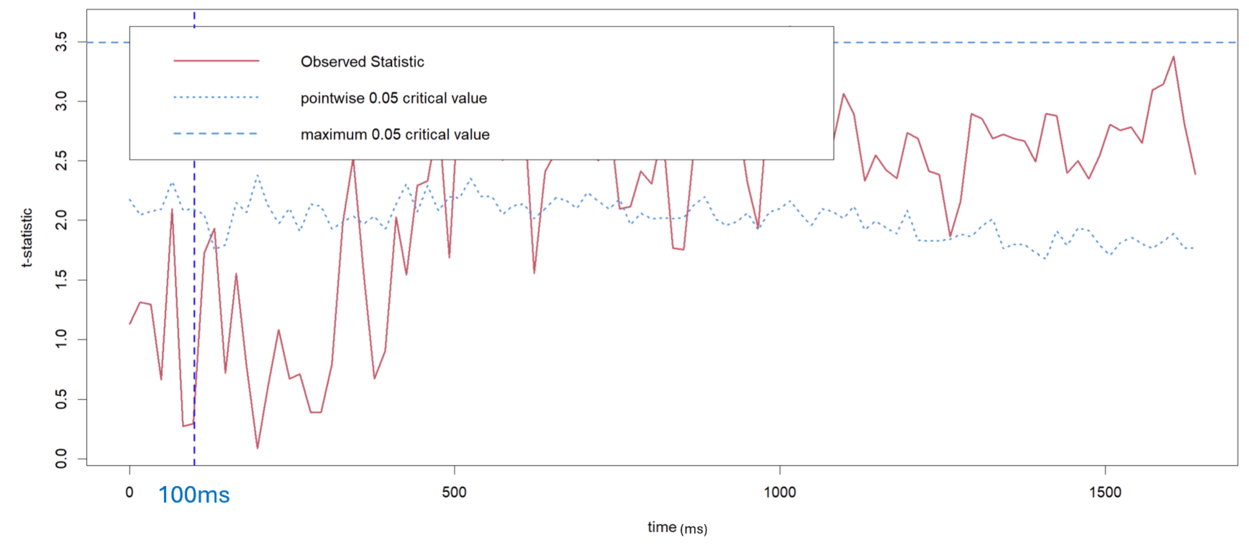

Figure 6: Permutation T Test Plot, Condition 1: Cases vs Controls

- Figure 6 shows an approximate significant difference between cases and controls following 400 ms.

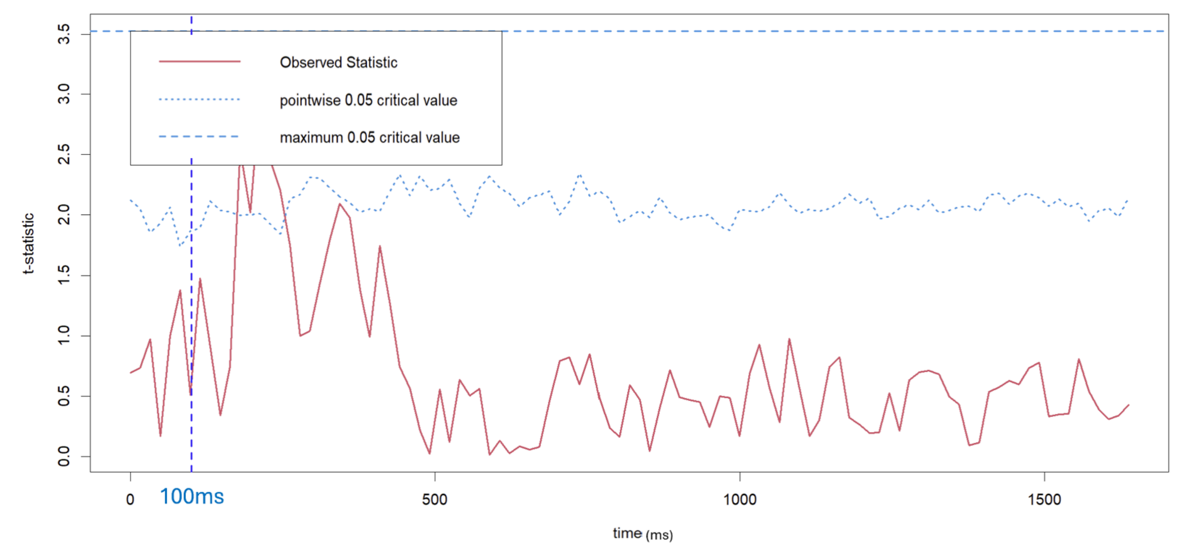

Figure 7: Permutation T Test Plot, Condition 2: Cases vs Controls

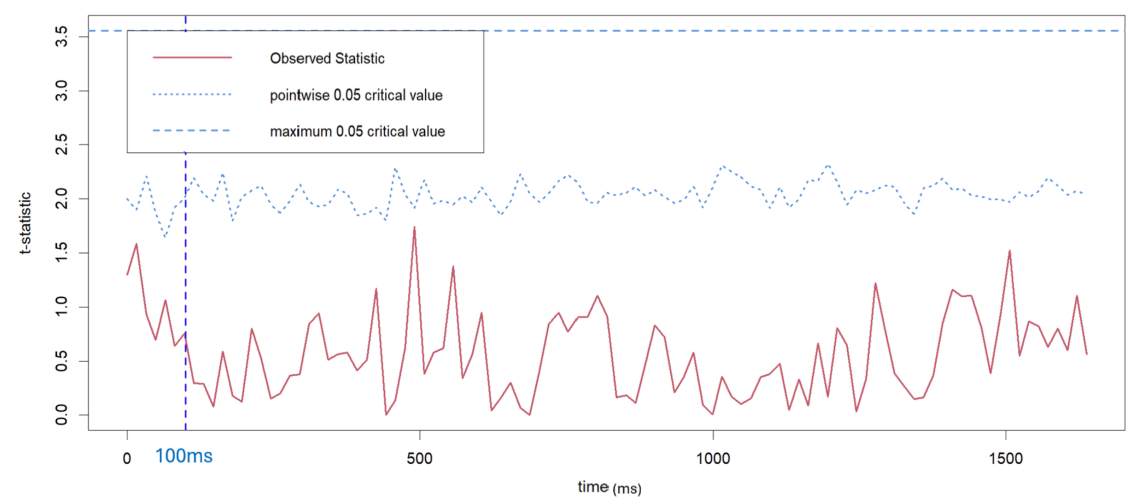

Figure 8: Permutation T Test Plot, Condition 3: Cases vs Controls

- Figures 7 and 8 show less significant differences for Conditions 2 and 3.

Conclusions

There is evidence to suggest our hypothesis based on the Figures produced by our t-test. This suggests that combined auditory and motor stimulation is required to trigger a significant EEG response in schizophrenia.

Next Steps

1. Test for population / sensor clustering
2. Repeated tests using other nodes e.g. FP2, FCZ
3. Functional Principal Component Analysis
4. Predictive modelling using features derived from FPCA scores

We will use the `fdapace` and `fda` packages [5] for this.

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: GitHub

References

1. Boutros et al., 2009, doi: https://doi.org/10.1016/j.schres.2009.05.019↩
2. Morozov 2023, Available at: https://www.istockphoto.com/vector/female-patient-do-eeg-scan-in-clinic-gm1474307021-504208695 (Accessed: 29 January 2026).↩
3. Tauscher 1998, doi: 10.1016/S0006-3223(97)00428-9↩
4. Kuo-Song Chang 2022, doi: 10.1371/journal.pone.0275870↩
5. Ramsay, J. O., Hooker, G., & Graves, S. (2009). Functional data analysis with R and MATLAB. Springer.↩